

K. J. Somaiya School of Engineering, Mumbai-77

Department of Electronics and Telecommunication Engineering

Name of the Laboratory: (Integrated Circuits) (A-213A)

Year: SY B.Tech **Branch:** EXTC **Semester:** III

Name of the Course: (Digital Logic Design Laboratory) (216U03L303)

Term: July 2025 to November 2025 (2025-2026)

Date: 14/07/2025

List of Experiments/Tutorials with CO and PO Mapping

Expt. No.	Name of experiment	Mapping with CO	Mapping with PO
1	1. Implementation of NOT, AND, OR, EX- OR gates using universal gates. 2. Design and Realization of Code converter using basic gates	CO1	PO1,PO2,PO5,PO8,PSO1
2	Study and realization of binary adder and binary subtractor using IC 7483	CO3	PO3,PO4,PO5,PO8,PSO1
3	Minimization and realization of logic function with standard and universal gates	CO1,CO2	PO2,PO3,PO5,PO8,PSO1
4	Study, design and implementation of logic functions with Multiplexer and Decoder	CO3	PO3,PO4,PO5,PO8,PSO1
5	Design and implementation of binary comparator	CO3	PO3,PO4,PO5,PO8,PSO1
6	Study and realization of S-R , J-K, T and D flip-flops and their conversions using VLAB and MSI ICs	CO4	PO3,PO4,PO5,PO8,PSO1
7	Design and implementation of Shift Registers using VLAB	CO4	PO3,PO4,PO5,PO8,PSO1
8	Study and implementation of different modes of shift registers using Universal Shift Register IC	CO4	PO3,PO4,PO5,PO8,PSO1
9	Design and implementation of Synchronous and Asynchronous counter using VLAB	CO4	PO3,PO4,PO5,PO8,PSO1
10	Study and implementation of R-2R D to A converter	CO5	PO3,PO4,PO5,PO8,PSO1

Note: For successful completion of term work minimum 10 number of experiment/tutorial needs to be submitted by student.

Name and Signature of Faculty Members:

Jyoti Varavadekar

Chaitali Kulkarni

Course Outcome	After Completion of the course student will be able to
CO1	Understand fundamentals of number systems, codes and arithmetic operations.
CO2	Minimize logic functions using different techniques and represent using logic gates.
CO3	Implement combinational logic circuits and their applications using various devices.
CO4	Design and implement sequential logic circuits and their applications.
CO5	Understand principle of data converters.

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Laboratory Assessment Rubrics

Parameter/Criteria	Poor (Marks)	Acceptable (Marks)	Good (Marks)
1)Design and Understanding	Incorrect design (0-3)	Correct design without understanding (4-5)	Correct design and clear understanding (6-7)
2)Lab work and Results	Many mistakes (0-3)	Needs help from others or faculty member (4-5)	Correct results with understanding (6-7)
3)Report writing	Incorrect / missing truth tables , diagrams ,conclusions (0-4)	Some details missing (5)	Neat and correct (6-7)
4)Timely submission	Late after 2 weeks (0-1)	Within 2 weeks (2)	Next practical turn (4)
For simulation experiments along with above 1,2 and 4 criteria			
Parameter			
Use of open source/licensed software	No (o marks)	Yes (2 marks)	-
Report writing	Screen shots prints/images missing (0-2marks)	Some details missing (3)	Neat and correct (4-5marks)

Name and Signature of Faculty Member:

Mrs Jyoti Varavadekar

Mrs Chaitali Kulkarni